

ORTONVILLE, MN, USA

WMO: 727515

Lat: 45.306N

Lon: 96.424W

Elev: 1100

StdP: 14.12

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 04982

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.6	-8.5	-19.6	1.9	-10.8	-16.6	2.3	-8.4	30.0	15.3	27.0	15.0	7.5	330	0.587

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	90.4	72.6	88.1	71.8	84.2	69.7	76.3	86.2	74.5	83.7	72.6	81.6	10.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.9	127.1	82.5	71.9	122.6	81.4	69.9	114.5	79.6	40.4	86.8	38.7	83.0	37.0	80.7	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.3	23.5	20.2	DB	-19.2	96.4	5.7	3.4	-23.3	98.8	-26.7	100.8	-29.9	102.8	-34.0	105.3	(4)
(5)			WB	-19.6	79.4	5.6	2.3	-23.7	81.1	-27.0	82.4	-30.1	83.8	-34.2	85.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.2	14.5	16.6	30.7	45.1	58.0	68.2	73.0	70.0	62.5	48.0	34.1	19.3		
	DBStd	23.07	13.42	12.89	13.09	10.60	8.81	6.38	5.61	5.56	8.11	9.49	11.32	12.37		
	HDD50	4480	1099	936	607	211	28	0	0	0	5	153	488	953		
	HDD65	7914	1564	1356	1062	598	251	39	7	16	141	534	928	1418		
	CDD50	2715	0	0	10	66	277	547	713	622	380	90	10	0		
	CDD65	674	0	0	0	2	35	136	256	173	66	6	0	0		
	CDH74	6234	0	0	4	59	415	1220	2346	1439	656	94	1	0		
	CDH80	2053	0	0	0	15	135	377	840	449	210	27	0	0		
	WSAvg	9.0	9.2	9.6	10.0	11.1	10.3	8.4	6.9	6.9	8.0	9.1	9.0	9.2		
	PrecAvg	25.0	0.7	0.7	1.3	2.5	2.8	3.9	3.9	2.7	2.8	2.3	1.0	0.6		
(15) Precipitation	PrecMax	32.2	1.7	1.6	3.7	6.8	5.1	7.5	7.9	5.3	6.8	6.8	3.8	1.8		
	PrecMin	15.9	0.0	0.1	0.2	0.3	0.9	1.3	1.6	0.6	0.3	0.2	0.0	0.1		
	PrecStd	4.3	0.5	0.4	0.8	1.4	1.0	1.7	1.7	1.4	1.5	1.8	1.0	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.7	52.1	71.7	81.5	90.2	91.2	95.3	93.1	90.7	83.9	68.5	50.0		
		MCWB	38.8	45.4	55.6	59.4	66.2	72.0	76.3	74.1	70.7	62.7	53.9	40.6		
	2%	DB	39.4	44.8	62.6	72.9	82.5	88.1	90.6	88.3	85.9	74.7	61.5	43.4		
		MCWB	35.0	39.0	51.8	54.4	63.3	70.3	74.7	73.5	68.2	58.8	49.6	36.9		
	5%	DB	36.5	38.9	54.4	68.1	78.9	84.0	88.1	84.2	81.5	68.1	54.8	38.7		
		MCWB	33.3	35.5	47.2	52.7	61.1	68.5	73.7	71.1	65.9	55.6	46.1	34.5		
	10%	DB	33.9	35.9	47.9	63.1	73.1	81.3	84.1	81.7	77.1	63.3	49.6	35.8		
		MCWB	31.8	33.5	43.0	50.3	58.5	66.6	71.4	69.2	64.5	52.7	43.3	33.1		
	0.4%	WB	39.7	46.0	58.7	61.4	70.6	76.1	79.9	77.8	74.4	66.2	56.7	42.0		
		MCDB	45.9	51.4	64.7	75.7	84.7	86.8	89.5	88.4	84.2	78.6	65.7	48.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.7	39.2	52.7	57.4	66.4	73.3	77.3	75.4	71.4	60.8	51.2	37.7		
		MCDB	39.6	43.5	61.0	69.4	78.0	83.5	87.5	84.7	81.7	71.7	59.7	42.4		
	5%	WB	33.5	36.0	47.3	54.3	63.8	71.0	75.4	73.4	69.0	57.1	46.8	35.0		
		MCDB	36.1	39.2	54.0	65.1	74.5	81.1	84.6	81.6	78.0	66.6	54.5	38.5		
	10%	WB	31.4	32.8	42.7	51.3	60.9	68.6	73.4	71.2	66.4	53.8	43.0	32.5		
		MCDB	33.5	35.5	48.2	61.1	70.8	78.0	82.1	79.3	74.5	61.4	49.5	35.1		
	5% DB	MDBR	16.5	17.1	16.9	19.4	20.0	18.8	19.4	19.5	21.1	18.7	17.3	14.8		
		MCDBR	20.1	20.3	26.4	29.9	27.8	24.2	22.8	23.2	26.8	28.4	26.7	20.9		
		MCWBR	16.4	15.8	17.0	15.3	13.0	11.2	11.0	11.3	12.2	15.2	16.7	15.8		
		MCWBR	18.5	19.4	25.6	25.5	23.2	20.1	20.4	20.7	22.4	24.3	25.4	18.5		
(40) Clear-Sky Solar Irradiance	taub		0.267	0.277	0.308	0.351	0.371	0.385	0.389	0.395	0.359	0.320	0.287	0.265		
		taud	2.429	2.417	2.440	2.359	2.374	2.389	2.365	2.331	2.417	2.502	2.507	2.447		
	Ebn at Noon		272	291	295	288	285	280	277	271	273	271	263	261		
		Edh at Noon	22	27	31	36	37	37	37	37	31	25	21	20		
(44) All-Sky Solar Radiation	RadAvg		560	904	1206	1529	1752	1965	2083	1739	1343	859	561	412		
	RadStd		61	97	111	121	124	118	81	147	94	113	58	41		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page